Explanation Stability and Post-hoc Trust Governance for Class-Incremental Network Traffic Classification

OFRA prediction with an offline ETG evidence ledger

WU LIQIANG

Abstract

Class-incremental network-traffic classification requires a model to learn newly introduced classes without an oracle task identifier while retaining earlier decisions. This paper studies prediction, explanation stability, and post-hoc governance as linked but separately bounded layers. OFRA combines a shared tabular encoder, family-specific low-rank binary heads, bounded replay, and a DP-Means router. Expected-Gradients SHAP explains the registered joint-cap class margin, and ETG records certification, refusal, escalation, and strict recertification outcomes without changing the prediction. Strict five-seed prediction evidence is available for NSL-KDD, UNSW-NB15, CIC-IDS-2017, and MalayaNetwork_GT, with model capacity disclosed per dataset. On Malaya FT512x12, joint cap 3,000 improves Macro-F1 by 9.37 percentage points and balanced accuracy by 7.97 points relative to head-only, while the accuracy difference is -1.34 points and the forgetting difference is +0.39 points. A source-bound Malaya seed-1 analysis identifies 12 silent-drift events among 17 eligible transitions and executes the ETG state machine on the same routed score. A three-method robustness pilot quantifies attribution dependence, while a no-look-ahead audit confirms Task-0-only numerical scaling but identifies bounded future-category schema exposure in NSL-KDD and UNSW-NB15. The evidence supports a reproducible offline audit pipeline and dataset-dependent trade-offs, not an online feedback loop or universal superiority.

The cumulative CatBoost and TabM rows retrain on all seen-class training samples and therefore have more information than OFRA's bounded replay pipeline. They diagnose classification headroom but cannot isolate the effect of the encoder, family heads, router, or memory budget. The CatBoost 400-tree setting is exploratory because it followed a seed-1 sensitivity check; no confirmatory superiority claim is made.

Keywords: class-incremental learning; network traffic classification; explainable AI; explanation drift; SHAP; trust governance; intrusion detection.

1. Introduction

1.1 Background and motivation

Network traffic changes as applications, services, attacker behaviour, and collection environments evolve. A fixed classifier therefore becomes incomplete, but repeatedly retraining one shared model can overwrite knowledge of earlier classes. Class-incremental learning addresses this setting by presenting new classes sequentially and evaluating all classes observed so far. For a security-facing system, predictive retention is only one requirement. Analysts may also rely on explanations to understand why a class score was produced, so a stable accuracy value does not by itself guarantee stable decision evidence.

This study distinguishes prediction, explanation, and governance. OFRA produces the class decision. SHAP explains the registered winning-class margin at stored checkpoints. ETG consumes performance and explanation evidence after inference and writes a governance ledger. This separation prevents an explanation-quality failure from silently cancelling an intrusion alert, while also avoiding the stronger claim that ETG already controls the learner.

1.2 Problem statement

Current class-incremental NIDS evaluation commonly prioritises accuracy and forgetting, whereas explanation stability and operational response are less consistently measured. A valid audit must explain the same score used by the registered inference path; an approximate wrapper or a separate surrogate model cannot establish governance of the deployed decision. The present study therefore uses a source-bound analysis of OFRA's registered joint-cap score. The remaining problem is to determine whether the current predictor retains useful performance across increments, whether its explanation ranking changes under performance-preserved transitions, and how uncertain or unstable explanations should be recorded without overstating deployment maturity.

1.3 Research questions and objectives

RQ1: Under legitimate class-incremental model updates, how stable are the feature attributions supporting OFRA's registered routed decision when class-level predictive performance remains within a declared stability rule?

RQ2: What accuracy, Macro-F1, balanced-accuracy, and forgetting trade-offs arise between registered OFRA decision arms, and what additional headroom is indicated by bounded classifier comparisons?

RQ3: How can explanation evidence be certified, refused, escalated, and re-certified after prediction without suppressing the underlying network-classification alert?

RO1 is to implement a fixed-probe, adjacent-checkpoint protocol for the actual joint-cap class margin. RO2 is to compare registered OFRA decision arms and bounded encoder or classifier diagnostics without changing the primary method or mixing unmatched protocols. RO3 is to operate an offline ETG ledger whose state transitions and limitations are explicitly auditable.

2. Related Work

2.1 Continual learning and parameter-efficient adaptation

Catastrophic forgetting motivates regularisation, rehearsal, distillation, exemplar, and parameter-isolation methods in continual learning (French, 1999; Kirkpatrick et al., 2017; Rebuffi et al., 2017). LoRA freezes a base transformation and represents a trainable update with low-rank matrices (Hu et al., 2022). For tabular representation learning, TabM uses parameter-efficient ensembling to produce multiple predictions within one model (Gorishniy et al., 2025), while CatBoost is a gradient-boosted tree method designed to reduce prediction shift in boosting (Prokhorenkova et al., 2018). The present family-specific low-rank head and TabM mean-embedding adapter are project integration choices; neither is presented as a new derivation of the cited methods.

2.2 Explanation drift and IDS governance

Recent work has moved beyond one-time post-hoc explanation. E2D2 detects explanation-based drift in unsupervised NIDS, but it targets distributional drift monitoring rather than legitimate class-incremental parameter updates (Nugraha and Bauschert, 2026). Maseno et al. (2026) use SHAP instability as an auxiliary signal for adversarial evasion, whereas the present study excludes adversarial perturbation and compares consecutive learning checkpoints. Goncalves and Alomari (2026) evaluate analyst-facing SHAP and LIME explanations in cloud intrusion detection, but do not define a class-incremental explanation ledger. Adabara et al. (2025) combine SHAP, probability calibration, and uncertainty in a cognitive IDS, yet do not test non-suppressing governance actions after explanation drift. These studies establish closely related motivation, but none closes the specific gap between legitimate class expansion, the exact routed decision score, and a governance record that leaves the predictive alert active.

Table 1. Positioning against the four recent studies identified during review.

Study	Drift or trust trigger	IDS setting	Gap relative to this work
E2D2 (2026)	Unsupervised explanation-based drift	Unsupervised NIDS	Not legitimate class-incremental model updates; no ETG action ledger

Maseno et al. (2026)	Adversarial SHAP instability	Recurrent multiclass IDS	Adversarial trigger, not benign learning increments; no non-suppressing governance
Goncalves and Alomari (2026)	Human interpretation of SHAP/LIME	AWS GuardDuty	User study rather than longitudinal routed-score stability
Adabara et al. (2025)	Calibration, uncertainty, and SHAP	CICIDS2017	No class-incremental sequence or explanation-state transition ledger

2.3 Research gap

The gap is not simply the absence of SHAP in NIDS. It is the absence of a source-bound protocol that (i) explains the same routed score used for prediction, (ii) measures stability under legitimate class-incremental updates, and (iii) converts that evidence into auditable, non-suppressing governance states. This paper addresses that narrower gap and avoids claiming that explanation stability alone proves prediction correctness or security effectiveness.

3. Methodology

3.1 Scope and dataset separation

Each dataset is preprocessed, trained, and evaluated independently; records from different datasets are never pooled. The current strict five-seed prediction table uses seeds 1, 2, 3, 4, and 42. NSL-KDD, UNSW-NB15, and CIC-IDS-2017 use FT-Transformer 256x4, while MalayaNetwork_GT uses FT-Transformer 512x12; these rows are therefore reported as dataset-specific evidence rather than a capacity-matched pooled comparison. Malaya is an application-traffic dataset rather than an intrusion-detection benchmark. A protocol-separated CIC-IDS-2018 FT256x4 closure also exists under a shorter 1+1-epoch schedule, while the stricter current CIC-IDS-2018 campaign remains in progress and is excluded from the headline table.

Dataset-specific preprocessing removes identifiers, normalises labels, converts features to finite numerical arrays, fixes feature and class order, and constructs class-by-split shards. NSL-KDD retains its official train/test files and produces 122 features; UNSW-NB15 likewise retains its official files and produces 194 features. Malaya uses 31 derived-flow CSVs at frozen revision 384a5927..., removes IP addresses, ports, and timestamps, and holds out one capture per class, producing 77 numerical features. The capture-level Malaya split avoids leakage from correlated rows within one capture.

The code/data no-look-ahead audit confirms that numerical standardisation reads only Task-0 training shards, freezes before pretraining, and rejects later updates. It also finds a bounded transductive categorical-schema limitation: relative to Task-0-only vocabularies, NSL-KDD has six future-only one-hot columns affecting 115 of 12,703 later-task rows (0.905%), and UNSW-NB15 has eight affecting 712 of 79,341 rows (0.897%). This is neither official-test leakage nor future-label training, but the affected preprocessing contracts are not strictly no-look-ahead. CIC-IDS-2017, CIC-IDS-2018, and Malaya have no data-derived categorical vocabulary.

3.2 OFRA architecture and registered decision arms

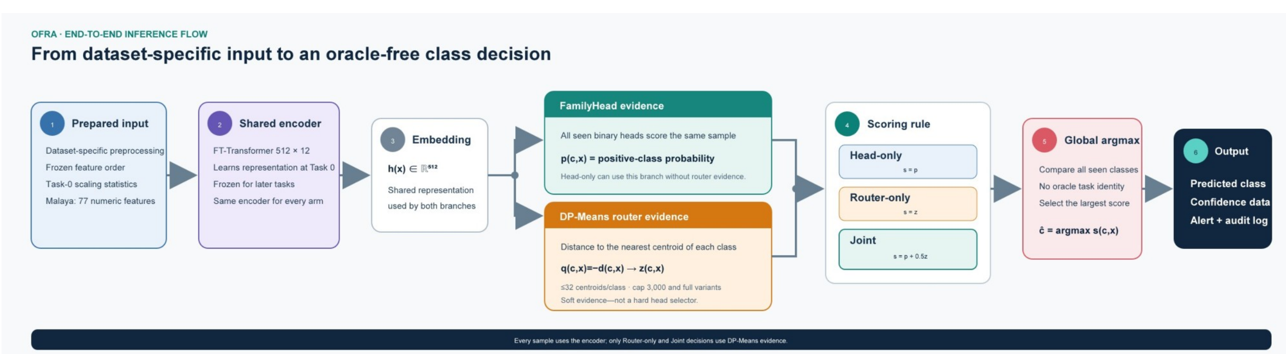

Figure 1. End-to-end OFRA inference. The DP-Means router supplies soft class evidence; it is not a hard selector of one family head.

The shared FT-Transformer maps a numerical row to a 512-dimensional representation using 12 layers, 16 attention heads, head dimension 32, and dropout 0.1. Task 0 is trained for eight epochs; each later task is trained for ten epochs. Each seen class has a rank-8, alpha-16 positive-versus-negative low-rank head. Replay retains at most 50 exemplars per class from a candidate pool of at most 5,000.

The DP-Means router stores up to 32 centroids per class. The capped router fits centroids from at most 3,000 selected training samples per class; the cap limits router fitting cost and majority-class influence, not test-set size. The registered arms are head-only p(c,x), router-only z(c,x), and joint p(c,x)+0.5z(c,x), each with capped and full router variants. Prediction is the global argmax across all seen classes. The joint weight 0.5, cap 3,000, distance quantile 0.3, and centroid limit 32 are study-defined registered settings rather than universal constants.

3.3 Training, metrics, and statistical boundary

The reported FT-Transformer configuration uses focal loss (gamma 2.0, alpha 0.75), learning rate 0.001, training batch size 384, evaluation batch size 512, and a negative-to-positive sampling ratio of four. At every checkpoint, evaluation covers all classes seen so far. Accuracy measures all correct predictions; Macro-F1 averages class F1 equally; balanced accuracy averages class recall; forgetting is the mean reduction from each earlier task's best historical accuracy to its later value. Current formal prediction values use seeds 1, 2, 3, 4, and 42. Model capacity and schedule are disclosed per dataset, and conclusions are descriptive unless a paired interval is explicitly reported.

3.4 Additive classifier-comparison protocol

TabM is added as a minimally invasive OFRA encoder comparison (Gorishniy et al., 2025). It produces 16 member embeddings of width 256 and averages them to the single representation required by the existing family-head interface. The family heads, exemplar budget, DP-Means routers, joint weight 0.5, cap 3,000, task order, and metric definitions remain unchanged. This is a mean-embedding adapter rather than a member-wise TabM-OFRA redesign.

CatBoost is evaluated separately as a cumulative multiclass diagnostic (Prokhorenkova et al., 2018). At each checkpoint it retrains on all training rows from classes seen so far. It does not use OFRA family heads, bounded replay, DP-Means routing, or ETG, and its native feature attribution is not the registered routed-margin SHAP target. The comparison therefore estimates classification headroom in the fixed feature representation rather than a matched continual-learning effect.

3.5 SHAP explanation protocol

The analyzer uses shap.GradientExplainer, an expected-gradients approximation for the differentiable FT-Transformer. It reconstructs the official joint-cap class margin g_c(x)=joint_cap_c(x)-max_{f not equal c} joint_cap_f(x) and attributes that margin to input features. Positive attributions support the explained margin; negative attributions oppose it. Top-15 features are ranked by absolute attribution magnitude. The fixed probe and background contracts are source-bound and reused across adjacent checkpoints.

For adjacent checkpoints, top-feature overlap is J(A,B)=|A intersection B|/|A union B|. The registered silent explanation-drift rule requires the class recall to fall by no more than 5 percentage points while top-15 Jaccard falls below 0.70. Both thresholds are project governance settings rather than externally validated standards. The event unit is a class-by-adjacent-checkpoint transition, not a packet, flow, sample, or real-world incident.

3.6 ETG governance scope

ETG is an offline explanation-governance ledger. It can certify an admission, refuse insufficient explanation evidence, escalate a drifted transition, and require strict recertification. A failed explanation action withholds the explanation alert or routes the evidence for review; it does not cancel OFRA's underlying classification alert. The completed analysis consumes stored OFRA checkpoint evidence for Malaya seed 1. ETG has therefore been connected to the actual routed-score evidence path, but it has not been evaluated as a closed loop that changes future routing, training, or replay.

3.7 Attribution-method robustness protocol

The source-bound robustness pilot uses the same Malaya seed-1 checkpoints, fixed probes, joint-cap-3000 class margin, top-15 rule, random-control admission threshold, and ETG state machine. Its three primary rankings are Expected Gradients, single-feature ablation to the frozen checkpoint mean, and Gradient x Input. Pairwise top-15 Jaccard, admission agreement, ETG-state agreement, and transition-level silent-drift agreement are reported. Integrated Gradients were also attempted with a Task-0 mean baseline and 16-point Gauss-Legendre quadrature, but its completeness check failed on the piecewise routed score; it is retained as a numerical diagnostic and excluded from primary agreement claims. This design measures method dependence rather than assuming that any explainer is uniquely correct.

3.8 No-look-ahead audit protocol

The audit traces every data-dependent preprocessing state to its fitting split and task availability. It checks numerical accumulator reads and freeze points, categorical-vocabulary construction, official-test access, and row-level impact of future-only categories. The audit distinguishes future-category schema exposure from future-label training and from test leakage, then reports exact affected columns and rows rather than a binary pass/fail label.

4. Results

4.1 RQ1: Explanation stability

In the completed Malaya seed-1 source-bound analysis, 17 class-by-adjacent-checkpoint transitions satisfy the eligibility rules and 12 meet the registered silent explanation-drift definition, giving 70.59%. This value must always be labelled as a single-seed Malaya result rather than a prevalence estimate, because the evidence unit is a registered class transition and not traffic volume or an operational incident.

The high event proportion shows that stable predictive evidence under the chosen tolerance does not guarantee stable top-feature membership. It does not establish that the model became unsafe, that the explanation was wrong, or that 70.59% of network flows drifted. A single-seed sensitivity grid over top-k {10, 15, 20}, Jaccard thresholds {0.6, 0.7, 0.8}, and allowed class-recall drops {2, 5, 10} percentage points confirms that the event rate is threshold-sensitive. The tolerance is class recall, not overall accuracy, and the grid does not establish a universal threshold.

4.2 RQ1: Attribution-method robustness

The three primary methods agree on admission for 40.0% of 30 checkpoint-class rows, on ETG state for 43.3%, and on the silent-drift conclusion for 20.0% of 20 adjacent class transitions. Pairwise mean top-15 Jaccard ranges from 0.380 to 0.567. Expected Gradients reports 12/17 silent-drift events, feature ablation 0/17, and Gradient x Input 7/17. The result bounds rather than removes attribution dependence: ETG outcomes are method-conditioned, and the registered Expected-Gradients result is not treated as uniquely correct.

Table 2. Attribution-method agreement on the source-bound Malaya seed-1 routed score.

Pair	Top-15 Jaccard	Admission	ETG state	Drift event
Expected vs ablation	0.567	56.7%	60.0%	40.0%
Expected vs Grad x Input	0.380	63.3%	60.0%	35.0%
Ablation vs Grad x Input	0.405	60.0%	56.7%	65.0%

Integrated Gradients are excluded from Table 2 because their completeness check failed (mean row-level error 11.58; maximum 373.72). They are retained as a failed numerical diagnostic. The three valid rankings quantify sensitivity but do not determine which explainer is causally correct.

4.3 RQ2: Predictive performance and retention

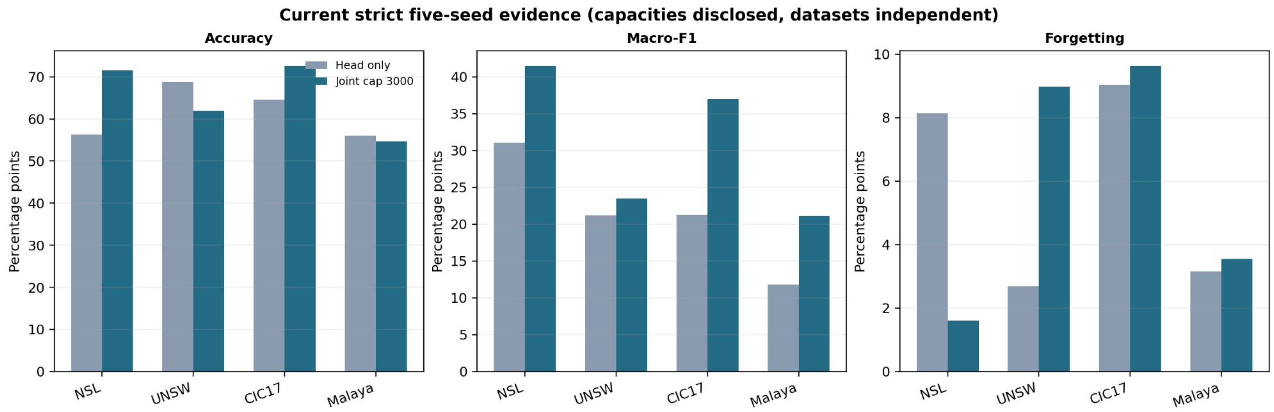

Figure 2. Head-only versus joint-cap current strict five-seed means. Capacities are disclosed by dataset and are not pooled.

Table 3. Current strict five-seed prediction evidence for the two main decision arms.

Dataset / model	Arm	Accuracy	Macro-F1	Balanced acc.	Forgetting
NSL-KDD FT256x4	Head only	56.29	31.08	36.48	8.14 pp
NSL-KDD FT256x4	Joint cap	71.56	41.46	42.25	1.60 pp
UNSW-NB15 FT256x4	Head only	68.78	21.16	23.52	2.68 pp
UNSW-NB15 FT256x4	Joint cap	61.95	23.50	28.45	8.97 pp
CIC-IDS-2017 FT256x4	Head only	64.59	21.25	25.91	9.04 pp
CIC-IDS-2017 FT256x4	Joint cap	72.57	36.95	63.13	9.63 pp
Malaya FT512x12	Head only	56.03	11.77	15.05	3.16 pp
Malaya FT512x12	Joint cap	54.68	21.15	23.03	3.55 pp

The routing effect is dataset-dependent. On NSL-KDD, joint cap 3,000 improves accuracy, Macro-F1, balanced accuracy, and forgetting. On UNSW-NB15, it improves Macro-F1 and balanced accuracy but lowers overall accuracy and increases forgetting. On CIC-IDS-2017, it improves accuracy and class-balanced metrics but does not reduce forgetting. On Malaya, the paired five-seed difference is -1.34 accuracy points, +9.37 Macro-F1 points, +7.97 balanced-accuracy points, and +0.39 forgetting points; pooled per-class results show non-zero recall for all ten classes under joint cap, but several minority recalls remain low. The table therefore supports dataset-specific trade-offs, not universal superiority.

A protocol-separated CIC-IDS-2018 FT256x4 campaign completed seeds 1, 2, 3, 4, and 42 with one Task-0 epoch and one epoch per later task. Head-only inference gives 69.27 +/- 20.34% accuracy, 21.81 +/- 8.82% Macro-F1, 25.67 +/- 6.35% balanced accuracy, and 7.62 +/- 7.54 points of forgetting. Joint cap-3000 gives 52.55 +/- 5.79%, 34.05 +/- 7.54%, 53.93 +/- 7.79%, and 19.03 +/- 8.62 points, respectively. This older closure remains valid only under its shorter schedule. The stricter current CIC-IDS-2018 campaign is still in progress and is not used in the current four-dataset table.

4.4 RQ2: Additive classifier comparison

Table 4. Malaya classifier comparisons. Values are mean +/- sample SD across five seeds; protocols remain separate.

Model and protocol	Accuracy	Macro-F1	Balanced acc.	Forgetting
TabM-OFRA joint cap, 5 seeds	57.99 +/- 3.93	20.46 +/- 1.19	21.60 +/- 1.70	4.39 +/- 2.56 pp
FT512-OFRA joint cap, 5 seeds	54.68 +/- 2.71	21.15 +/- 3.37	23.03 +/- 3.42	3.55 +/- 0.77 pp
CatBoost cumulative, 5 seeds	66.50 +/- 0.44	34.76 +/- 0.69	39.10 +/- 0.73	15.34 +/- 0.73 pp
TabM cumulative, 5 seeds	65.01 +/- 0.96	32.66 +/- 1.31	34.41 +/- 1.51	9.43 +/- 1.80 pp

Across the same five seeds, TabM-OFRA joint-cap accuracy is 57.99% versus 54.68% for FT512-OFRA, a +3.31-point difference. Macro-F1 is 20.46% versus 21.15% (-0.69 points), balanced accuracy is 21.60% versus 23.03% (-1.43 points), and forgetting is 4.39 versus 3.55 points (+0.84 points). The TabM change therefore improves aggregate accuracy but not the class-balanced or retention metrics. Because BitTorrent is the majority class, this result is not evidence of uniform minority-class improvement.

4.5 RQ3: ETG governance outcomes

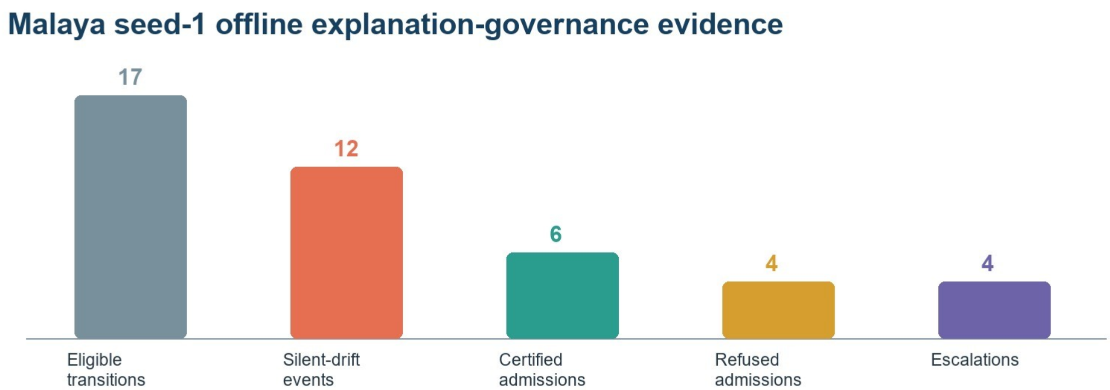

Figure 3. Malaya seed-1 ETG evidence. Counts are governance outcomes, not production incidents or completed human reviews.

ETG records six certified admissions and four refused admissions. Four transitions are escalated, one strict recertification succeeds, and two strict-recertification attempts fail. Final states include six UNEXPLAINABLE, three CERTIFIED_STABLE, and one DRIFTED class record. These outcomes demonstrate execution of the registered state machine on actual joint-cap explanation evidence. They do not quantify analyst benefit, false-alarm reduction, production remediation, or causal improvement of later OFRA checkpoints.

5. Discussion

5.1 When is the trade-off operationally worthwhile?

The value of routing depends on the dataset and operational cost model. NSL-KDD is favourable across the four displayed metrics. UNSW-NB15 trades lower accuracy and higher forgetting for better class-balanced metrics. CIC-IDS-2017 gains predictive coverage without a retention gain. Malaya improves coverage of minority application classes while its accuracy and forgetting differences remain uncertain. Such a trade may be worthwhile when a missed underrepresented class is more costly than a small aggregate-accuracy loss, but the decision must be based on per-class costs and cannot be inferred from overall accuracy alone.

5.2 Classifier capacity and class imbalance

The new comparisons separate two questions that were previously conflated. TabM-OFRA asks whether a different differentiable encoder can improve the existing OFRA pipeline without changing its router, heads, or memory rules. The cumulative models ask how much performance is available when all seen-class rows are retained and a fresh multiclass classifier is trained. TabM-OFRA improves aggregate accuracy but not class-balanced metrics, whereas cumulative CatBoost improves all final classification metrics while showing greater checkpoint forgetting. Together with the per-class recall audit, this indicates that model capacity alone is not the main bottleneck; class imbalance and the combination of binary heads, routing, and bounded replay remain material.

5.3 Meaning of the integrated evidence chain

The current contribution is an integrated offline audit chain: OFRA generates the registered joint-cap decision; expected gradients explain its class margin; ETG applies declared evidence rules and records the outcome. Calling this a closed-loop OFRA-ETG system would be premature because ETG does not alter later model behaviour. The distinction matters for both scientific validity and system safety: post-hoc governance evidence can support review without being presented as a validated adaptive control mechanism.

5.4 Attribution dependence

The completed source-bound pilot confirms that ETG states inherit material attribution-method dependence. Agreement must therefore be reported at several levels rather than reduced to one feature-overlap number. Expected Gradients remains the registered operational explainer, but governance conclusions should be treated as method-conditioned; a future deployment should require method consensus, uncertainty-aware admission, or external calibration. The failed Integrated-Gradients completeness check further shows that a differentiable attribution method cannot be assumed valid for the complete piecewise routed score.

6. Limitations and Threats to Validity

First, strict five-seed prediction evidence is complete for four datasets, but the current stricter CIC-IDS-2018 campaign remains in progress and model capacity differs between Malaya and the three FT256x4 datasets. Second, SHAP and ETG evidence remains Malaya seed 1; additional seeds are not yet complete. Third, attribution dependence is bounded by a three-method single-seed pilot rather than a multi-seed estimate, and Integrated Gradients failed its completeness diagnostic. Fourth, silent-drift thresholds are study-defined operating rules and have not been externally calibrated. Fifth, ETG actions are simulated and no analyst study or deployed feedback loop has been conducted.

Sixth, numerical Task-0-only scaling passes the no-look-ahead audit, but NSL-KDD and UNSW-NB15 categorical schemas contain bounded future-category information affecting about 0.9% of later-task rows; strict vocabulary rebuilding and rerunning remain open. Seventh, Malaya is not an intrusion-detection dataset, and its application labels cannot support attack-recall or benign false-positive claims. Eighth, Malaya is strongly imbalanced and several minority recalls remain low even after routing. Ninth, cumulative CatBoost and TabM diagnostics are not matched continual-learning baselines. Tenth, complete runtime, explanation latency, energy, analyst burden, and protocol-matched external-baseline tests are not yet available across the full dataset suite.

7. Conclusion

This revision provides a bounded evaluation of OFRA prediction, additive classifier diagnostics, explanation stability, and post-hoc ETG governance. Strict five-seed prediction evidence across NSL-KDD, UNSW-NB15, CIC-IDS-2017, and MalayaNetwork_GT shows that routing effects are dataset-dependent. On Malaya, joint cap improves class-balanced metrics but does not establish an accuracy or forgetting gain. A source-bound Malaya seed-1 analysis finds 12 silent explanation-drift events among 17 eligible transitions and executes the ETG state machine on the same routed-score evidence; threshold sensitivity and attribution-method dependence are reported explicitly. The work supports a reproducible integrated offline audit pipeline, not an online adaptive loop or universal superiority claim. Remaining submission work is multi-seed explanation/ETG analysis, the strict categorical-vocabulary rebuild and rerun, the current CIC-IDS-2018 closure, and stronger protocol-matched continual-learning baselines.

Declarations

Funding: No external funding was declared for this study.

Conflict of interest: The author declares no conflict of interest.

Ethics and data governance: The study uses previously collected public benchmark or research datasets and does not involve direct interaction with human participants. Dataset licences and institutional computing rules remain applicable. No restricted credentials or private institutional data are included in the public repository.

Code and data availability: Reproducibility code, configurations, preprocessing contracts, result summaries, and public evidence bindings are available at https://github.com/1838904818/Class-IL. Original datasets remain with their respective providers; the repository documents acquisition and preparation rather than redistributing data where licence or size makes redistribution inappropriate.

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